

Role of Information Technology in the Environment

Technological Advancements: a bane or boon?

Technology has caused many environmental and social problems, but it is also key to addressing environmental degradation, climate change, food scarcity, waste management, and other pressing global challenges. A recent rise in global concern for climate change has led to the development of new environmental technology aiming to solve some of the biggest environmental concerns that we face as a society, by transitioning towards a more sustainable, low-carbon economy. Environmental technology is also known as 'green' or 'clean' technology and refers to the development of new technologies which aim to conserve, monitor or reduce the negative impact of technology on the environment and the consumption of resources.



This section will focus on the role of information technology in improving the fields of environmental education and human health. The emerging growth of internet services and facilities, geographic information system or GIS, and the data that gets transmitted through satellites have generated a wealth of updated information on several aspects of the environment, as well as health. Let us look at some ways in which technology is helping the environment.

1) Advancements in Energy Efficiency

The advancement of technology has made us more energy-efficient in terms of electricity consumption. Slowly, but surely, our electrical applications have become more efficient, and these incremental changes over a period of time have a dramatic effect in the long run. For example, a modern LED light is around 60% more efficient than traditional incandescent bulbs and has a lifespan that's 50 times longer. Now, we don't need to make and constantly discard more lights. This is a huge step in reducing waste, using fewer resources and less energy; so LED bulbs are a small but important part of tackling our environmental troubles.

2) Role of Technology in Saving Wildlife

Technology is being used widely in saving wildlife. Some of the measures taken worldwide in protecting the fauna are:

- 1) Smart collars embedded with GPS, meters, and sensors to keep track of endangered species like rhinos and elephants.
- 2) Remote monitoring of wildlife sounds and noises to detect any predator harm or natural distress.
- 3) SIM-based collars for animals near human habitats to reduce animal-human conflicts.
- 4) Gene sequencing techniques to save endangered species from incurable diseases like cancer.



- 5) Conservation drones to track and monitor wild forest regions for any natural disasters like forest fires that can cause animals to be killed. 6) Predictive analytics for animals to gather information about every species on the planet and work comprehensively in protecting them.

3) Greener Energy Production

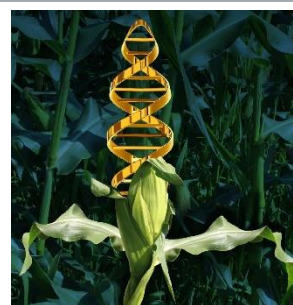


The process of energy production is becoming cleaner. As we learn to harness the power of the sun, wind and water, our dependence on power sources that are exhaustible and produce pollution is reduced year by year. Renewable energies made up 25% of total global energy production in 2017 and this number is predicted

to be bumped up to 85% by the year 2050. Clean energy systems themselves are becoming more efficient. For example, in 1960, the maximum efficiency we could achieve from solar cells was 16%. Recently, a group of scientists in the US announced they were able to achieve 44.5% efficiency.

4) Energy-Efficient Farming

Our ways of farming are becoming less harmful to the environment as we begin to understand how our actions affect surrounding ecosystems. Genetically modified crops are able to improve yields dramatically, and as we better understand how to get the most out of our crops, we need less land to produce the same amount of food. Meat grown definitely raises a few raised eyebrows, but it is predicted to become a valuable source of cleaner, greener protein for humans. It can also be tailored to our specific nutritional needs and cut out harmful compounds.



5) Improvements in Recycling Technology

The next important contribution is the improvement in recycling technology. In the future, humans may be able to create a 100% closed loop system by recycling the majority of the materials we use. This would be one of the most significant advancements for the planet as the lifecycle of something as simple as a water bottle can lead to resource depletion, industrial waste, air and water pollution, habitat erosion, animal extinction and more.

6) Cleaner Technologies that Prevent Pollution

Advancements in technology have helped prevent an incredible amount of harm to our planet towards the end of the last century. For example, inventions such as the catalytic converter, found in most cars, have prevented billions of tons of toxic pollutants from entering the atmosphere. Similar industrial systems, such as filters and other converters, have also helped reduce pollutants. Shifting towards electric vehicles will also remove a huge percentage of greenhouse gas emissions.

As people have greater access to information, more people can realize their impact and find ways to combat it. For example, the hole created by CFCs in the ozone was only discovered due to the

invention of the Dobson Ozone Spectrophotometer, which allowed us to understand the full extent of damage these gases were having. Artificial intelligence (AI) is another advancement that could transform our perspective of the environment. It could help us to better understand the long-term effects of our actions through simulations and predictions.

Apart from the obvious advancements in technology that have led to the betterment of the environment, there exist databases and certain other management systems that assimilate information on the various aspects of the environment.

7) Databases on the Environment

So, what are Databases on the environment: Database is a collection of connected data on certain subjects. It comes in a computerized form, and can be retrieved at any hour of the day, whenever necessary. Some of the available databases include

- Wildlife databases,
- Forest cover database, and
- Conservation database, etc. They are also available on
- Diseases including malaria, fluorosis, HIV/AIDS, etc.

Some of the commonly used databases include

1. National management information system (NMIS).
2. Environmental information system (ENVIS).
3. Remote sensing and geographical information system (GIS), and last and not the least,
4. The World Wide Web (WWW).

National Management Information System (NMIS)

NMIS is a division under the Department of Science and Technology (DST, India) that is responsible for the collection, collation, analysis and dissemination of information on resources devoted to science & technology activities in the country. NMIS maintains a close collaboration with UNESCO, NSF, OECD, etc. for matters relating to S&T statistics and indicators. The next database is the

Environment Information System (ENVIS)

ENVIS is a one stop, web-enabled, comprehensive portal which provides information on the environment and related subject areas to researchers, academicians, policy planners, environmentalists, scientists, engineers and the general public. It is a decentralized network of databases in operation since 1982 and is run by the Ministry of Environment Forests and Climate Change. ENVIS serves as the backbone of decision making and environment management at all levels of Government and it has been making valuable contribution in environmental protection and its improvement. Some of the information available in ENVIS include topics such as:

1. Pollution control
2. Clean technologies
3. Remote sensing
4. Coastal ecology
5. Biodiversity
6. Western and eastern Ghats

7. Environmental management
8. Media related to environment, renewable energy, desertification, mangroves, wildlife, Himalayan ecology, mining, etc.

The most important benefit of technology in the environment is probably remote sensing and GIS or geographic information systems.

Geographical information system (GIS)

GIS is a computer-based tool for mapping and analyzing feature events on earth and is integrated with maps. Any location on the earth has massive amounts of data, including physical features, but political, economic and social data, as well. GIS facilitates the process by which we can visualize, analyze and understand this data. GIS is an effective tool for

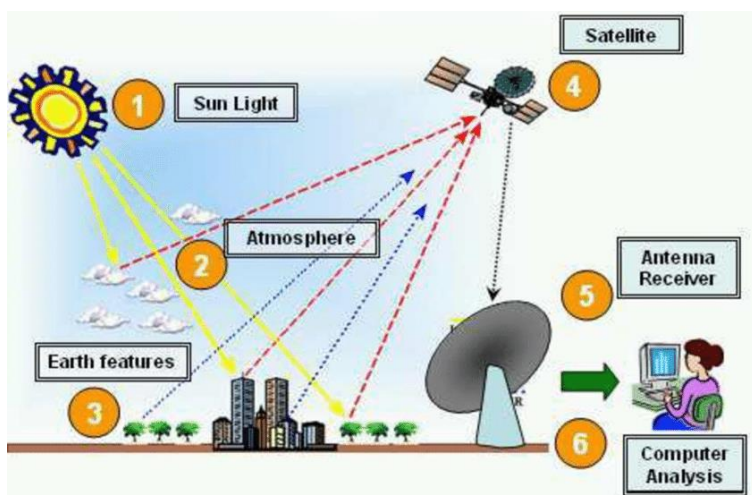


- a) Studying the environment
- b) Reporting on environmental phenomena
- c) Modeling how the environment responds to natural and man-made factors

GIS can be also be used to visualize data about natural resources, hazards, pollution emissions, health of ecosystems, climate change and so on. It is also very effective tool to analyze ecological footprints; improve watershed resource management; and respond to climate change, pollution, and much more. Now, let us look at remote sensing.

Remote Sensing

Remote sensing is one of the methods commonly used for collecting physical data to be integrated into GIS. Remote sensors collect data from objects on the earth without any direct contact. They do this by detecting energy reflected from the earth; these detectors are typically mounted on satellites or aircraft. Remote sensing technology has become much more prevalent, accurate and accessible in the recent years, and covers a wide range of engineering applications. For example, Satellite imageries provide us with real-time information about various physical and biological resources, water logging, desertification, deforestation, urban sprawl, river and canal network, mineral and energy reserves and so on.



Digital information on a number of aspects like water resources, industrial growth, human settlements, road network, soil type, forest land, crop land or grassland can be accessed and analyzed. Data on monsoons, ozone layer depletion, smog can help in taking steps to deal with these issues effectively. Several softwares are also available that help along in this quest for the improvement of the environment.

Satellite Imaging

Anyone with an internet connection can now use Google Earth software to monitor or support scientific research on almost any place on the earth's surface. High-resolution satellite imagery is used to produce images using this software. Several ocean monitoring satellites have also been launched recently, which provide additional information required on various data such as a) Surface winds, ocean surface strata, b) Chlorophyll concentrations in ocean waters, c) Monitoring phytoplankton blooms, d) Atmospheric aerosols and suspended sediments in water. Several projects have been launched by the government of India, for the betterment of human health with the aid of technology. Health SAT is one such initiative by ISRO (Indian Space Research Organization).

HealthSAT

HealthSAT is a telemedicine system, housed in a small health centre. It consisting of a PC with customized medical software connected to a few medical diagnostic instruments, such as an ECG or X-ray scanner for scanning X-ray photos. Digitized versions of patients' medical images and diagnostic details (such as X-ray and blood reports) are dispatched to specialist doctors through the satellite-based communication link. The information, is in turn, received at the specialist centre where experienced doctors examine these reports, diagnose, interact with the patients (along with local doctors), and suggest appropriate treatment through video-conferencing.

Kiosk

The Information Kiosk, on the other hand, is an IT based facility available in rural areas. It provides information on groundwater use and cropping system changes. This technology can also be made use of in the conservation of wildlife and for the protection of endangered species. The Kiosk software is a touch screen driven setup with (no physical keyboard) with large icons displaying limited but focused data supported by graphics and animation. Local language is for ease of understanding, though it can be accessed in multiple languages, with simple and clear navigational paths.

Software Monitoring of Wildlife

In a bid to save the endangered big cats or tigers, the Centre has introduced M-Stripes, a software monitoring system to strengthen effectiveness of surveillance and anti-poaching measures in all the 39 tiger reserves across the country. This program has been reported to have achieved tremendous success and has witnessed a decline in poaching of these big cats.